SEQUENCES, SERIES AND DIFFERENTIAL EQUATIONS

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What we have learned in the previous lecture

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- First order non-homogeneous differential equations
  - Methods of integrating factors
  - Method of variation of parameters
- Second order differential equations
  - Equations reconducible to first order equations

Today's agenda

- Linear second order difference equations with constant coefficients
- Non-homogeneous second order linear equations
- Laplace transform
  - Solving differential equations with Laplace transform

Linear differential equations with constant coefficients

$$a\,y'' + b\,y' + cy = 0,$$

- We search a solution in the form $y=e^{rt}$ .Y will be a solution if and only if

$$ar^2 + br + c = 0, \Rightarrow \quad r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = -\frac{b}{2a} \pm \frac{\sqrt{D}}{2a},$$

- Case 1

$$D = b^2 - 4ac > 0. \quad r_1 = \frac{-b - \sqrt{D}}{2a}, \qquad r_2 = \frac{-b + \sqrt{D}}{2a}.$$

$$y = y_1(t) = e^{r_1 t} \qquad y_2(t) = e^{r_2 t} \qquad y = A\,e^{r_1 t} + B\,e^{r_2 t}$$

- Case 2

$$D = b^2 - 4ac = 0. \Rightarrow \quad r_1 = r_2 = -b/(2a) = r$$

$$y = A\,e^{rt} + Bt\,e^{rt}.$$

Linear differential equations with constant coefficients

- Case 3

$$D = b^2 - 4ac < 0. \Rightarrow r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = k \pm i\omega,$$

$$y = A\,e^{kt}\cos(\omega t) + B\,e^{kt}\sin(\omega t).$$

Nonhomogeneous second order linear equations

$$a_2(x)\frac{d^2y}{dx^2} + a_1(x)\frac{dy}{dx} + a_0(x)y = f(x).$$

- We assume that two independent solutions $y_1(x)$ and $y_2(x)$ of the corresponding homogenenous system are known. The general solution can be then expressed as

$$y = y_p(x) + y_h(x) = y_p(x) + C_1 y_1(x) + C_2 y_2(x),$$

- How to calculate $y_p(x)$?

Non-homogeneous second order linear equations

Method 1: undertermined coefficients → make an educated guess!

EXAMPLE 1 Find the general solution of $y'' + y' - 2y = 4x$.

Solution Because the nonhomogeneous term $f(x) = 4x$ is a first-degree polynomial, we "guess" that a particular solution can be found that is also such a polynomial. Thus, we try

$$y = Ax + B, \qquad y' = A, \qquad y'' = 0.$$

Substituting these expressions into the given DE, we obtain

$$0 + A - 2(Ax + B) = 4x \qquad \text{or}$$
$$-(2A + 4)x + (A - 2B) = 0.$$

This latter equation will be satisfied for all x provided $2A + 4 = 0$ and $A - 2B = 0$. Thus, we require $A = -2$ and $B = -1$; a particular solution of the given DE is

$$y_p(x) = -2x - 1.$$

Since the corresponding homogeneous equation $y'' + y' - 2y = 0$ has auxiliary equation $r^2 + r - 2 = 0$ with roots $r = 1$ and $r = -2$, the given DE has the general solution

$$y = y_p(x) + C_1e^x + C_2e^{-2x} = -2x - 1 + C_1e^x + C_2e^{-2x}.$$

Non-homogeneous second order linear equations

- Method 2: variation of parameters

$$y_p = u_1(x)y_1(x) + u_2(x)y_2(x).$$

$$u_1' = -\frac{y_2(x)}{W(x)}\frac{f(x)}{a_2(x)}, \qquad u_2' = \frac{y_1(x)}{W(x)}\frac{f(x)}{a_2(x)}, \qquad W(x) = \begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix}.$$

Laplace transform

F is of **exponential order** if it is at least piecewise continuous on the interval $[0, \infty)$ and satisfies

$$|F(t)| \leq Ke^{-ct}, \qquad \text{for some positive constants } K \text{ and } c.$$

The Laplace Transform

If $F(t)$ is of exponential order, the Laplace transform $L\{F(t)\}$ is defined by

$$L\{F(t)\} = \int_0^\infty e^{-st} F(t)\, dt = f(s).$$

Note that $f(s)$ exists at least for all $s > c$, the constant in the exponential order condition.

If $F(t)$ is of exponential order and is n times differentiable, and if $L\{F(t)\} = f(s)$, then

$$L\{F^{(n)}(t)\} = s^n f(s) - \left[s^{n-1} F(0) + s^{n-2} F'(0) + \cdots + F^{(n-1)}(0)\right].$$

Laplace transform

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EXAMPLE 1 If k is a constant, show that the Laplace transform of e^{kt} is

$$L\{e^{kt}\} = \frac{1}{s-k},$$

and use the result to solve the initial-value problem $y' - y = 0$, $y(0) = 1$.

Solution We have

$$L\{e^{kt}\} = \int_0^\infty e^{-st}\, e^{kt}\, dt = \int_0^\infty e^{(k-s)t}\, dt = \left.\frac{e^{(k-s)t}}{k-s}\right|_0^\infty = \frac{1}{s-k},$$

provided $s > k$. Since $y' - y = 0$, linearity of Laplace transforms implies that $L\{y'\} - L\{y\} = L\{0\} = 0$. Hence, by Theorem 5 we have $0 = sL\{y\} - y(0) - L\{y\} = (s-1)L\{y\} - 1$, and so

$$L\{y\} = \frac{1}{s-1} = L\{e^t\}.$$

By Theorem 4, the solution of the initial-value problem is $y = e^t$.

Laplace transform

$F(t) = L^{-1}\{f(s)\}$	$f(s) = L\{F(t)\}$	$F(t) = L^{-1}\{f(s)\}$	$f(s) = L\{F(t)\}$
1	$\frac{1}{s}, \quad (s > 0)$	e^{at}	$\frac{1}{s-a}, \quad (s > a)$
$t^n, n \geq 0$	$\frac{n!}{s^{n+1}}, \quad (s > 0)$	$t^p, \quad p > -1$	$\frac{\Gamma(p+1)}{s^{p+1}}, \quad (s > 0)$
$\sin(bt)$	$\frac{b}{s^2+b^2}, \quad (s > 0)$	$\cos(bt)$	$\frac{s}{s^2+b^2}, \quad (s > 0)$
$\sinh(at)$	$\frac{a}{s^2-a^2}, \quad (s > \lvert a \rvert)$	$\cosh(at)$	$\frac{s}{s^2-a^2}, \quad (s > \lvert a \rvert)$
$e^{at}\sin(bt)$	$\frac{b}{(s-a)^2+b^2}, \quad (s > a)$	$e^{at}\cos(bt)$	$\frac{s-a}{(s-a)^2+b^2}, \quad (s > a)$
$t\sin(bt)$	$\frac{2bs}{(s^2+b^2)^2}, \quad (s > 0)$	$t\cos(bt)$	$\frac{s^2-b^2}{(s^2+b^2)}, \quad (s > 0)$
$t^n F(t)$	$(-1)^n f^{(n)}(s)$	$F^{(n)}(t)$	$s^n f(s) - \sum_{j=0}^{n-1} s^{n-1-j} F^{(j)}(0)$
$H(t-c)$	$\frac{e^{-cs}}{s}$	$\delta(t-c)$	e^{-cs}
$H(t-c)F(t-c)$	$e^{-cs} f(s)$	$F * G(t) = \int_0^t F(t-\tau)g(\tau), d\tau$	$f(s)g(s)$
$I_1(t) = \int_0^t F(\tau)\, d\tau$	$\frac{1}{s} f(s)$	$I_n(s) = \int_0^1 I_{n-1}(s)$	$\frac{1}{s^n} f(s)$

Laplace transform

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EXAMPLE 5 **The Shifting Principle** Find the Laplace transform of $e^{at}F(t)$ in terms of $f(s) = L\{F(t)\}$. What does this imply for the inverse transform L^{-1}?

Solution Observe that

$$L\{e^{at}F(t)\} = \int_0^\infty e^{-st}e^{at}F(t)dt = \int_0^\infty e^{-(s-a)t}F(t)dt = f(s-a).$$

Thus, multiplying a function by an exponential shifts its Laplace transform. Therefore,

$$L\{e^{at}F(t)\} = f(s-a) \quad \text{and} \quad L^{-1}\{f(s-a)\} = e^{at}F(t).$$

Laplace transform

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- Convolution product of functions F(t) and G(t)

$$F * G(t) = \int_0^t F(t-u)G(u)\,du.$$

The Laplace transform of a convolution of two functions is the product of their Laplace transforms. If $L\{F(t)\} = f(s)$ and $L\{G(t)\} = g(s)$, then

$$L\{F * G\} = f(s)g(s).$$

It follows that $L^{-1}\{f(s)g(s)\} = F * G(t)$.

Differential equations -recap

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- First order differential equations
  - Separable equations
  - Non-homogenous equations
    - Method of integrating factors
    - Method of variation of parameters
- Second order differential equations
  - Equations reconducible to first order
  - Homogeneous equations with constant coefficients
  - Non-homogeneous equations
    - Method of undetermined coefficients
    - Method of variation of parameters
- Using Laplace transform for solving differential equations